

EEE4114F

Model Evaluation: Overfitting and Underfitting



Generalization

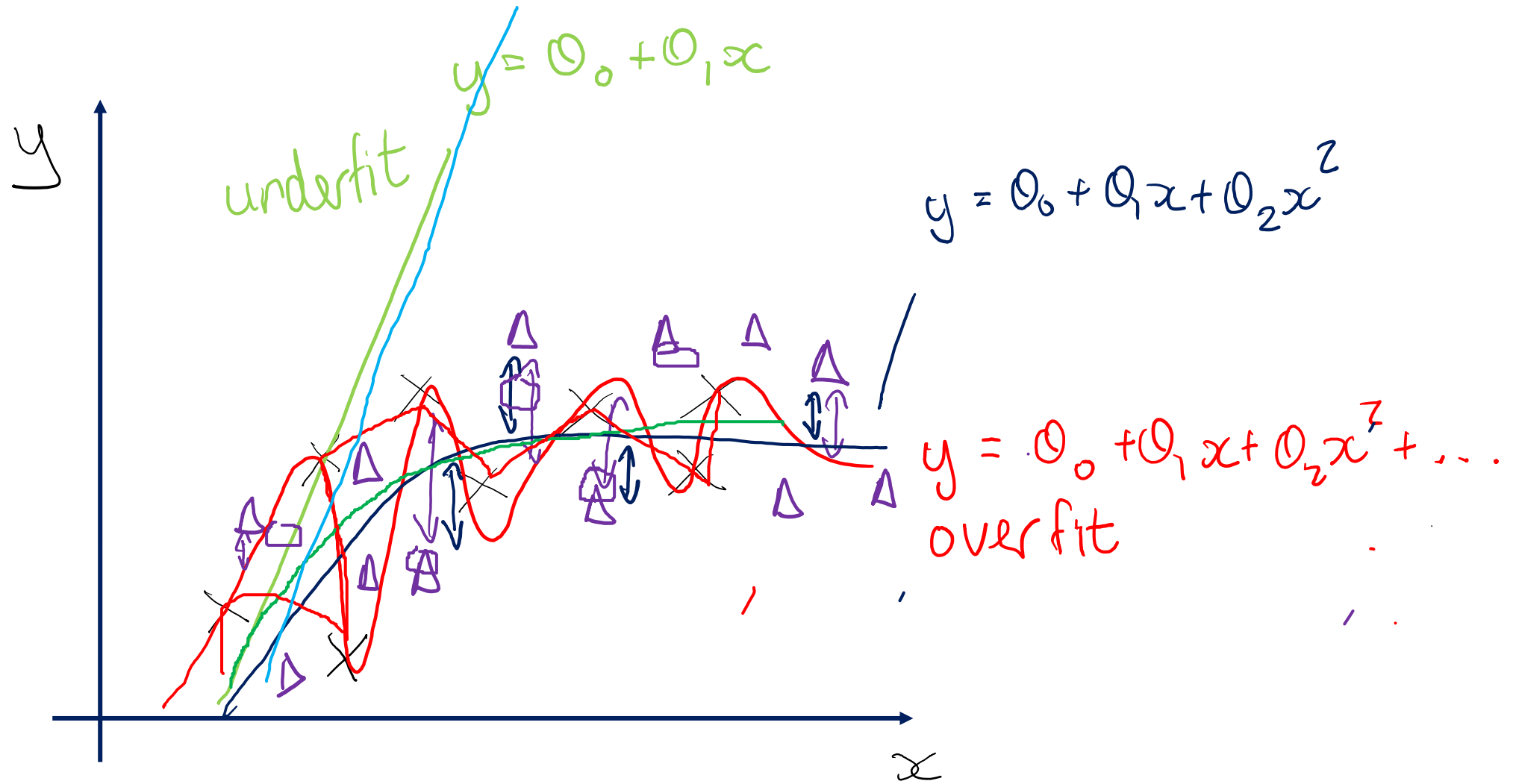
- How well does our model perform on unseen data
- Can we diagnose specific problems?
 - Overfitting, underfitting
 - Bias, variance
- Once diagnosed can we find appropriate solutions?

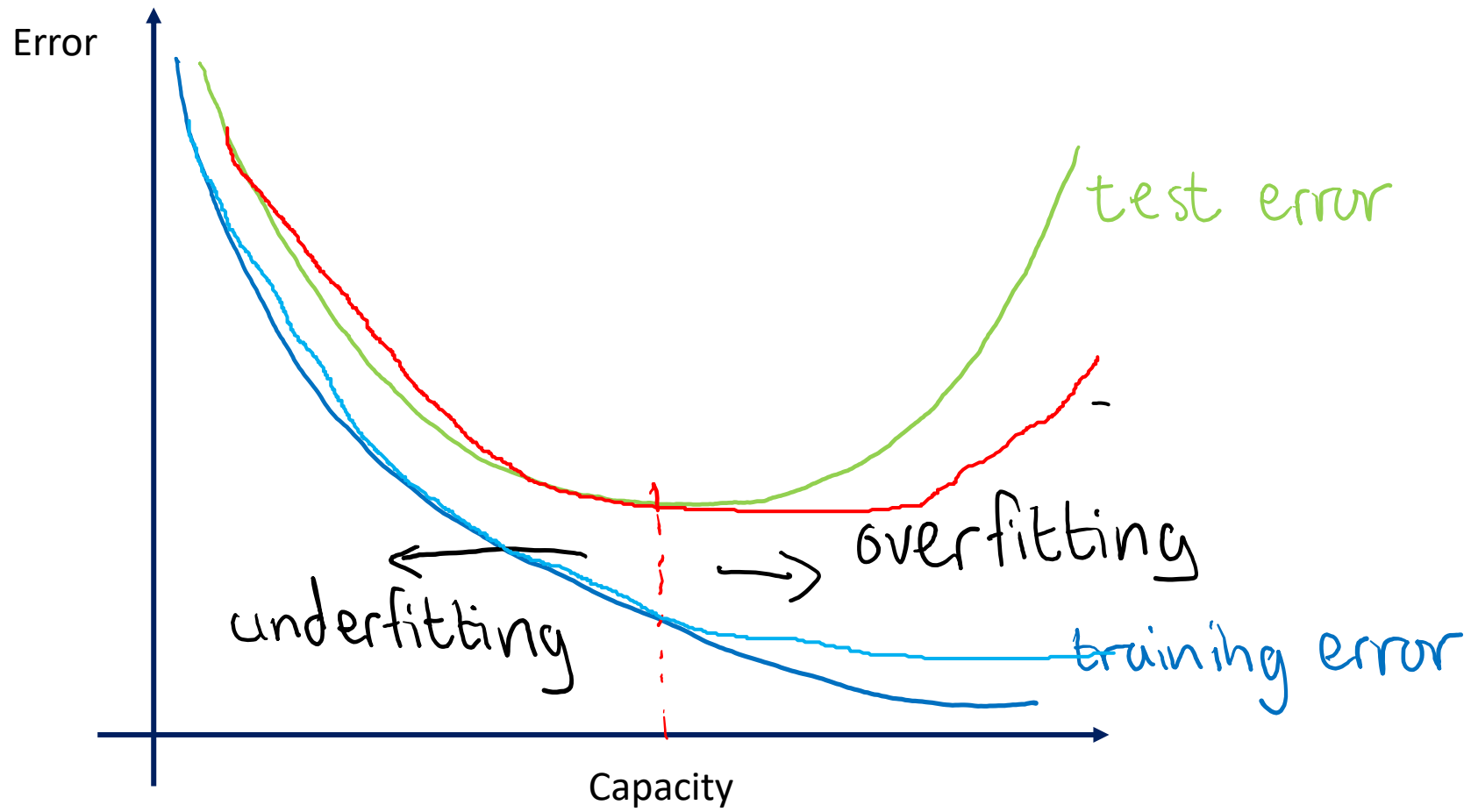


Overfitting and Underfitting

- Overfitting
 - Very common problem
 - Good performance on seen data but poor on unseen data
- Underfitting
 - Poor performance on both seen and unseen data







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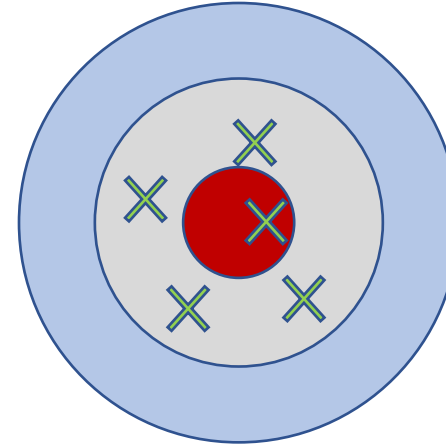
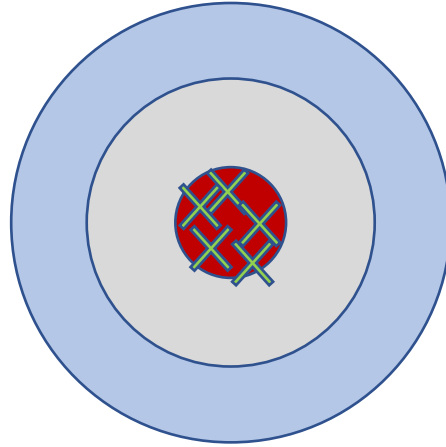
Bias and Variance Trade-off



Low variance

High variance

Low bias



High bias

